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Lecture 5

Density Estimation and Gaussian Mixture Models

Introducing Maximum Likelihood Estimation, and
Gaussian Mixture Models

EECS 189/289, Fall 2026 @ UC Berkeley

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Recap

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In the last lecture we introduced basic probability rules and derived the Maximum Likelihood Estimate (MLE) and started to derive the Maximum a Posteriori (MAP) Estimate.

Today we will:

- Finish deriving the MAP estimate
- Introduce the **Multivariate Gaussian** and how to **compute Bias**
- Introduce the **Mixture of Gaussians model** that combines many of these ideas.

Estimating the Parameters of a Distribution

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Given a collection of data $\mathcal{D} = \{x_1, \ldots, x_N\}$, there are several ways to estimate the "best" parameters for a given distribution.

- **Maximum Likelihood Estimation (MLE):** choose the parameters that maximize the **likelihood** of the data. (Frequentist)

$$w^*_{\text{MLE}} = \arg\max_w p(\mathcal{D} \mid w)$$

- **Maximum A Posteriori (MAP) Estimation:** choose the parameters that maximize the posterior given the data and the **prior**. (Bayesian)

$$w^*_{\text{MAP}} = \arg\max_w p(\mathcal{D} \mid w)\, p(w)$$

Both methods rely on computing the **likelihood of the data.**

The Bernoulli MLE Is Just Counting

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The maximum likelihood estimate (MLE) is:

$$\mu_{ML} = \frac{n_1}{n_0 + n_1} = \frac{n_1}{N}$$

This is **the fraction of ones in the data**, which is the **empirical distribution** defined earlier in lecture.

- μ_{ML} is the parameter value that makes the **observed data most probable** under a Bernoulli model.

The Parameter as a Random Variable

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Maximum likelihood estimation treats **$\boldsymbol{\mu}$ as an unknown constant.**

The Bayesian view treats **$\boldsymbol{\mu}$ as a random variable** with a distribution. Applying Bayes' theorem to the parameter:

$$p(\mu \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mu)\, p(\mu)}{p(\mathcal{D})}$$

- **Prior** $p(\mu)$: what we believe about μ before seeing data.
- **Likelihood** $p(\mathcal{D} \mid \mu)$: is the quantity we maximized for the MLE.
- **Posterior** $p(\mu \mid \mathcal{D})$: the updated belief after seeing the data.

Maximum a Posteriori (MAP) estimate maximizes the posterior:

$$\mu_{\mathrm{MAP}} = \arg\max_{\mu} \ln p(\mathcal{D} \mid \mu) + \ln p(\mu)$$

- The denominator $p(\mathcal{D})$ does not depend on $\boldsymbol{\mu}$, so it does not affect the maximization.

Modeling the Prior

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The **choice** and **parameterization** of model for the prior $p(\mu)$ is a **design decision**.

- Capture our prior beliefs about the μ.
- Model is often chosen for convenience (conjugate to the likelihood).
- The prior parameters are set using *prior knowledge* or *another prior*.

What would be a good prior $p(\mu)$ for the Bernoulli coin flip example?

- The parameter $0 \leq \mu \leq 1$ is the probability of landing heads
- We expect the coin (and the flipper) to be approximately fair.

The Beta Distribution

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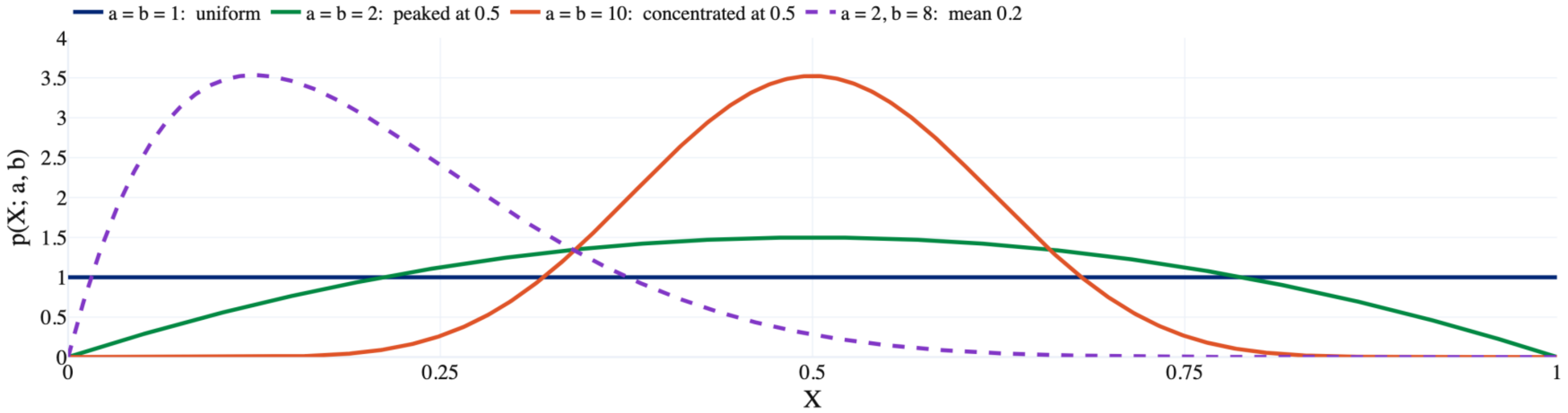
The **Beta distribution** for the **continuous random** variable $0 \le X \le 1$:

$$p(X; a, b) = \frac{1}{B(a,b)} X^{a-1}(1-X)^{b-1}$$

with **parameters by $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$** with the following shape:

- ➢$a = b = 1$: every value of X is equally plausible.
- ➢$a = b = 2$: values of X close to 0.5 are more likely
- ➢Large a and b concentrate the distribution near $\frac{a}{a+b}$

The beta function $B(a, b)$ is the normalizer (computed numerically) and outside the scope of this class.



Deriving the Posterior for Bernoulli + Beta

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Recall we are trying to compute:

$$p(\mu \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mu)\, p(\mu)}{p(\mathcal{D})}$$

Substituting the Bernoulli and the Beta distributions (without normalization):

$$p(\mu \mid \mathcal{D}) \propto p(\mathcal{D} \mid \mu)\, p(\mu) = \mu^{n_1}(1-\mu)^{n_0}\, \mu^{(a-1)}(1-\mu)^{(b-1)}$$

Collecting terms, we can see that the **posterior is also a Beta distribution**:

$$p(\mu \mid \mathcal{D}) \propto \mu^{(n_1+a-1)}(1-\mu)^{(n_0+b-1)}$$

with parameters $a' = (n_1 + a)$ and $b' = (n_0 + b)$.

When a prior paired with a likelihood has the same form as the posterior it is called a **conjugate prior**.

➢The Beta distribution is the conjugate prior for the Bernoulli likelihood.

Computing the MAP

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Maximum a Posteriori (MAP) estimate maximizes the posterior:

$$\mu_{\text{MAP}} = \arg\max_{\mu} \ln p(\mathcal{D} \mid \mu) + \ln p(\mu)$$

When the deriving the posterior we already showed (collecting terms):

$$\ln p(\mathcal{D} \mid \mu) + \ln p(\mu) = (n_1 + a - 1) \ln \mu + (n_0 + b - 1) \ln(1 - \mu)$$

When computing the MLE we derived the **stationary point** of:

$$\frac{n_1}{n_1 + n_0} = \arg\max_{\mu} n_1 \ln \mu + n_0 \ln(1 - \mu)$$

Substituting $n_1' = (n_1 + a - 1)$ and $n_0' = (n_0 + b - 1)$.

$$\mu_{\text{MAP}} = \frac{n_1 + a - 1}{n_1 + n_0 + a + b - 2} = \frac{n_1 + a - 1}{N + a + b - 2}$$

The Prior as Pseudo-Counts

The prior acts like a form of pseudo-counts (assumed binary observations) by contributing $a - 1$ imaginary successes $b - 1$ imaginary failures.

$$\mu_{\text{MAP}} = \frac{n_1 + a - 1}{n_1 + n_0 + a + b - 2} = \frac{n_1 + a - 1}{N + a + b - 2}$$

With $a = b = 2$ this becomes $\mu_{\text{MAP}} = \frac{n_1+1}{N+2}$ one imaginary observation of each outcome.

- Regardless of the observations $0 < \mu_{\text{MAP}} = \frac{n_1+1}{N+2} < 1$ the probability is never 0 or 1.

This is also called **Laplace Smoothing**

- Add **pseudo counts to empirical estimates** to act as a prior when there is **limited data** or **events are rare**.

How μ_{MAP} and μ_{ML} scale with N

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What happens as N grows and how do μ_{MAP} and μ_{ML} compare?

$$\mu_{ML} = \frac{n_1}{n_0+n_1} = \frac{n_1}{N}, \qquad \mu_{\text{MAP}} = \frac{n_1+a-1}{n_1+n_0+a+b-2} = \frac{n_1+a-1}{N+a+b-2}$$

As N grows, the constants $a-1$ and $a+b-2$ become negligible and the **two estimates converge**.

- **Small N:** the prior dominates, and the estimate is pulled toward the prior mean.
- **Large N:** the data dominates, and MAP and MLE agree.

Today many deep learning applications operate in the Large N (Big Data) setting where the MLE method is preferred (simpler).

- In future lectures, we will discuss introduce **regularization methods** that are equivalent or related to MAP estimation with priors.

For a,b > 2 which has higher variance?

More Practice with MLE

Starting Lecture 5

Normal (Gaussian) Distribution

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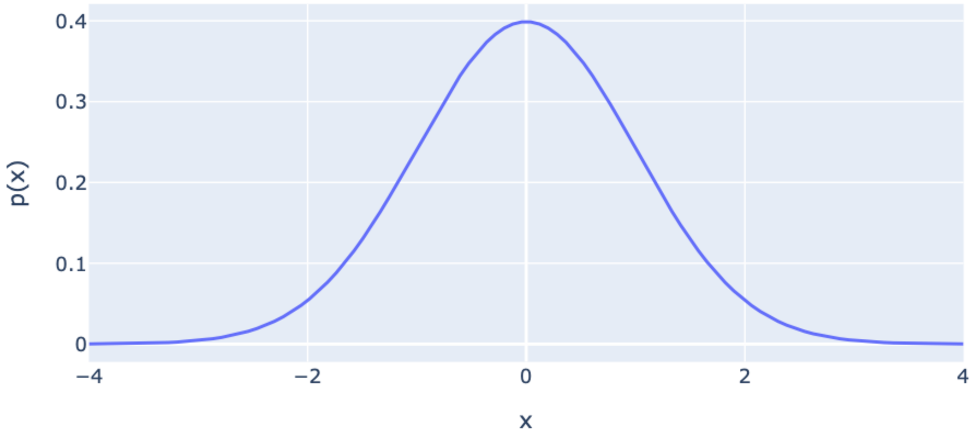
If a single real-valued variable X is **normally distributed,** then:

$$X \sim \mathcal{N}(\mu, \sigma^2) \quad \Longrightarrow \quad p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\,\pi\,\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

Where the parameters μ and σ^2 are called the mean and variance.



With some calculus you can show $\mathbb{E}[X] = \mu$ and $\mathrm{var}[X] = \sigma^2$.

The MLE for IID Gaussian Samples (Part 1)

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<u>Example</u>: Gaussian Distribution

We observe N continuous real valued samples $\mathcal{D} = \{x_1, \ldots, x_N\}$

1. **Modeling:** Let's assume IID Gaussian trials:

$$\ln p(\mathcal{D}|\mu, \sigma^2) = \sum_{n=1}^{N} \ln\left(\frac{1}{\sqrt{2\,\pi\,\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_n - \mu)^2\right)\right)$$

$$= -\frac{N}{2}\ln(2\,\pi) - \frac{N}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2$$

2. **Optimization:** Maximize the log likelihood function:

$$\arg\max_{\mu,\sigma^2} -\frac{N}{2}\ln(2\,\pi) - \frac{N}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2$$

The MLE for IID Gaussian Samples (Part 2)

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Maximize the log likelihood function:

$$\arg\max_{\mu,\sigma^2} -\frac{N}{2}\ln(2\,\pi) - \frac{N}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2$$

Compute derivative with respect to μ and σ^2 and solve for the critical-points.

$$\frac{\partial}{\partial\mu}\ln p(\mathcal{D}|\mu,\sigma^2) = \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n - \mu)$$

Set derivative equal to 0 and solve for μ_{ML} :

$$\frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n - \mu) = 0 \;\; \Longrightarrow \;\; \mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

The MLE for IID Gaussian Samples (Part 3)

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Computing the derivative with respect to the variance parameter $v = \sigma^2$

$$\frac{\partial}{\partial v} \ln p(\mathcal{D}|\mu, v) = \frac{\partial}{\partial v}\left(-\frac{N}{2}\ln(v) - \frac{1}{2v}\sum_{n=1}^{N}(x_n - \mu)^2\right) = -\frac{N}{2}\left(\frac{1}{v}\right) + \frac{1}{2v^2}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2$$

Set equal to zero and solving for $v_{\mathrm{ML}} = \sigma_{\mathrm{ML}}^2$:

$$-\frac{N}{2}\left(\frac{1}{v}\right) + \frac{1}{2v^2}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2 = 0$$

$$\frac{v^2}{N} * \quad N\left(\frac{1}{v}\right) = \frac{1}{v^2}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2 \quad * \frac{v^2}{N}$$

$$v_{\mathrm{ML}} = \sigma_{\mathrm{ML}}^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2$$

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Set equal to zero and solving for $v_{\mathrm{ML}} = \sigma_{\mathrm{ML}}^2$:

$$-\frac{N}{2}\left(\frac{1}{v}\right) + \frac{1}{2v^2}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2 = 0$$

$$\frac{v^2}{N} * \quad N\left(\frac{1}{v}\right) = \frac{1}{v^2}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2 \quad * \frac{v^2}{N}$$

$$v_{\mathrm{ML}} = \sigma_{\mathrm{ML}}^2 = \boxed{\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2}$$

$$\sigma_{\mathrm{ML}}^2 = \frac{1}{N}\sum_{n=1}^{N}\left(x_n^2 - 2\, x_n\, \mu_{\mathrm{ML}} + \mu_{\mathrm{ML}}^2\right) = \left(\frac{1}{N}\sum_{n=1}^{N} x_n^2\right) - \left(2\,\mu_{\mathrm{ML}}\frac{1}{N}\sum_{n=1}^{N} x_n\right) + \left(\frac{N\,\mu_{\mathrm{ML}}^2}{N}\right)$$

$$= \left(\frac{1}{N}\sum_{n=1}^{N} x_n^2\right) - \left(2\mu_{\mathrm{ML}}^2\right) + \left(\mu_{\mathrm{ML}}^2\right) = \boxed{\left(\frac{1}{N}\sum_{n=1}^{N} x_n^2\right) - \mu_{\mathrm{ML}}^2}$$

Recap of the MLE for IID Gaussian Samples

We used the method of **Maximum Likelihood** to estimate the **mean** and **variance parameters** of a Gaussian from data:

$$\mu_{\text{ML}} = \frac{1}{N} \boxed{\sum_{n=1}^{N} x_n}$$

Sufficient Statistics

$$\sigma^2_{\text{ML}} = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{\text{ML}})^2 = \frac{1}{N} \left(\boxed{\sum_{n=1}^{N} x_n^2} \right) - \mu^2_{\text{ML}}$$

What are the sufficient statistics?

Do they match our intuition for the mean and variance?

Are they good estimates?

Bias

- MAP Estimation
- Gaussians
  - Maximum Likelihood
  - Bias
- Gaussian Mixture Models

Questions

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Bias of an Estimate

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How close is the true (unknown) parameter value to our estimate in expectation?

Before we observe our data, the value of our estimate w_{ML} is a **random variable**.

We can compute the bias of the estimation procedure as:

$$\text{Bias}(w_{\text{ML}}) = \mathbb{E}[w_{\text{ML}}] - w$$

Relative to the true (but unknown) w.

Bias of μ_{ML} for the Gaussian Distribution

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Assuming the data are drawn IID from $\mathcal{N}(x|\mu,\sigma^2)$ treating X_n, μ_{ML},

$$\mathbb{E}[\mu_{\mathrm{ML}}] = \mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N} X_n\right] = \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[X_n] = \frac{1}{N}\sum_{n=1}^{N}\mu = \boxed{\mu} \quad \textbf{Unbiased}$$

This means that in expectation over randomly sampled datasets our **Maximum Likelihood estimate of the mean (μ_{ML})** will match the mean μ of the distribution $\mathcal{N}(x|\mu,\sigma^2)$ from which we sampled.

Is the maximum likelihood variance estimate biased?

Bias of σ^2_{ML} for the Gaussian Distribution

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Assuming the data are drawn IID from $\mathcal{N}(x|\mu,\sigma^2)$ treating X_n, μ_{ML}, and σ^2_{ML} as random variables

$$\mathbb{E}[\sigma^2_{\mathrm{ML}}] = \mathbb{E}\left[\left(\frac{1}{N}\sum_{n=1}^{N} X_n^2\right) - \mu^2_{\mathrm{ML}}\right] = \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[X_n^2] - \mathbb{E}[\mu^2_{\mathrm{ML}}]$$

$$= \frac{1}{N}\sum_{n=1}^{N}(\sigma^2+\mu^2) - \left(\boxed{\mathrm{var}(\mu_{\mathrm{ML}})} + \mathbb{E}[\mu_{\mathrm{ML}}]^2\right)$$

Variance Identity

$\mathrm{var}(X_n) = \mathbb{E}[X_n^2] - \mu^2 = \sigma^2$

$\mathrm{var}(\mu_{\mathrm{ML}}) = \mathbb{E}[\mu^2_{\mathrm{ML}}] - \mathbb{E}[\mu_{\mathrm{ML}}]^2$

$$\mathrm{var}(\mu_{\mathrm{ML}}) = \mathrm{var}\left(\frac{1}{N}\sum_{n=1}^{N} X_n\right) = \underbrace{\frac{1}{N^2}\left(\sum_{n=1}^{N}\mathrm{var}(X_n)\right)}_{\text{Independence of } X_n} = \frac{1}{N^2}N\underbrace{\sigma^2}_{\mathrm{var}(X_n)} = \frac{\sigma^2}{N}$$

Bias of σ^2_{ML} for the Gaussian Distribution

Assuming the data are drawn IID from $\mathcal{N}(x|\mu,\sigma^2)$ treating X_n, μ_{ML}, and σ^2_{ML} as random variables

$$\mathbb{E}[\sigma^2_{\mathrm{ML}}] = \mathbb{E}\left[\left(\frac{1}{N}\sum_{n=1}^{N} X_n^2\right) - \mu^2_{\mathrm{ML}}\right] = \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[X_n^2] - \mathbb{E}[\mu^2_{\mathrm{ML}}]$$

$$= \frac{1}{N}\sum_{n=1}^{N}(\sigma^2+\mu^2) - \left(\boxed{\mathrm{var}(\mu_{\mathrm{ML}})} + \mathbb{E}[\mu_{\mathrm{ML}}]^2\right)$$

$$= (\sigma^2+\mu^2) - \left(\frac{\sigma^2}{N} + \mu^2\right) = \sigma^2\left(\frac{N-1}{N}\right)$$

Variance Identity

$$\mathrm{var}(X_n) = \mathbb{E}[X_n^2] - \mu^2 = \sigma^2$$
$$\mathrm{var}(\mu_{\mathrm{ML}}) = \mathbb{E}[\mu^2_{\mathrm{ML}}] - \mathbb{E}[\mu_{\mathrm{ML}}]^2$$

Sample Mean Variance

$$\mathrm{var}(\mu_{\mathrm{ML}}) = \frac{\sigma^2}{N}$$

Bias of σ^2_{ML} for the Gaussian Distribution

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We have just shown that σ^2_{ML} **is biased** by a factor of $\left(\frac{N-1}{N}\right)$.

$$\mathbb{E}\left[\sigma^2_{\mathrm{ML}}\right] = \sigma^2\left(\frac{N-1}{N}\right)$$

How can we make σ^2_{ML} unbiased?

Correcting the Bias of σ^2_{ML}

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We showed that σ^2_{ML} is biased ($\mathbb{E}[\sigma^2_{\text{ML}}] \neq \sigma^2$):

$$\mathbb{E}[\sigma^2_{\text{ML}}] = \sigma^2 \left(\frac{N-1}{N}\right)$$

Sometimes called the **unbiased sample variance.**

We can correct the bias by multiplying by $\left(\frac{N}{N-1}\right)$:

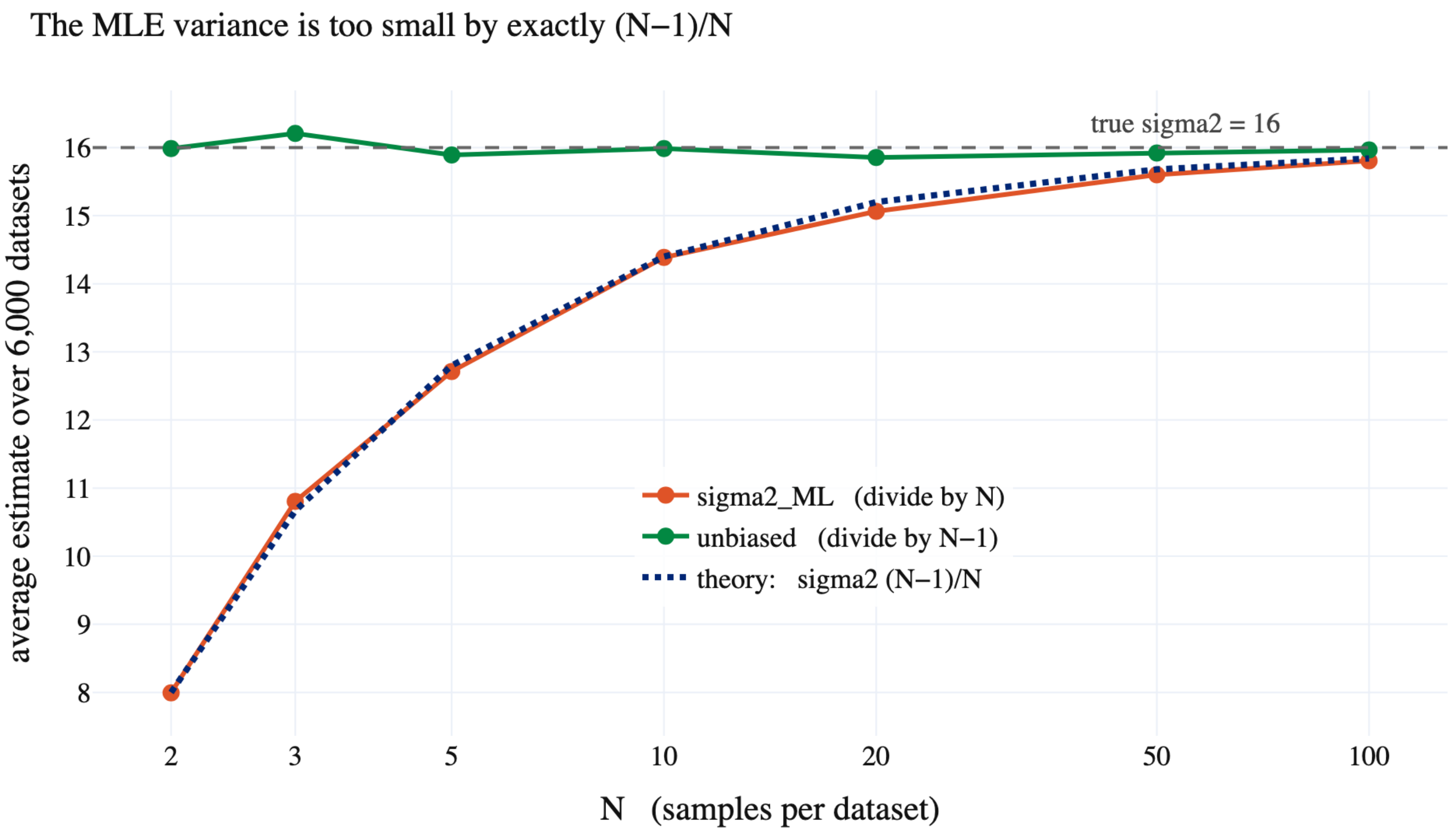
$$\left(\frac{N}{N-1}\right)\sigma^2_{\text{ML}} = \left(\frac{N}{N-1}\right)\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu_{\text{ML}})^2 = \frac{1}{N-1}\sum_{n=1}^{N}(x_n - \mu_{\text{ML}})^2$$

For large N the bias vanishes ➔ σ^2_{ML} is **asymptotically unbiased.**

- In future lectures, we will show that there is a tradeoff between bias and variance and we may prefer the biased estimator.

Demo

Why np.var has a ddof argument



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Fitting Data with Multiple Modes

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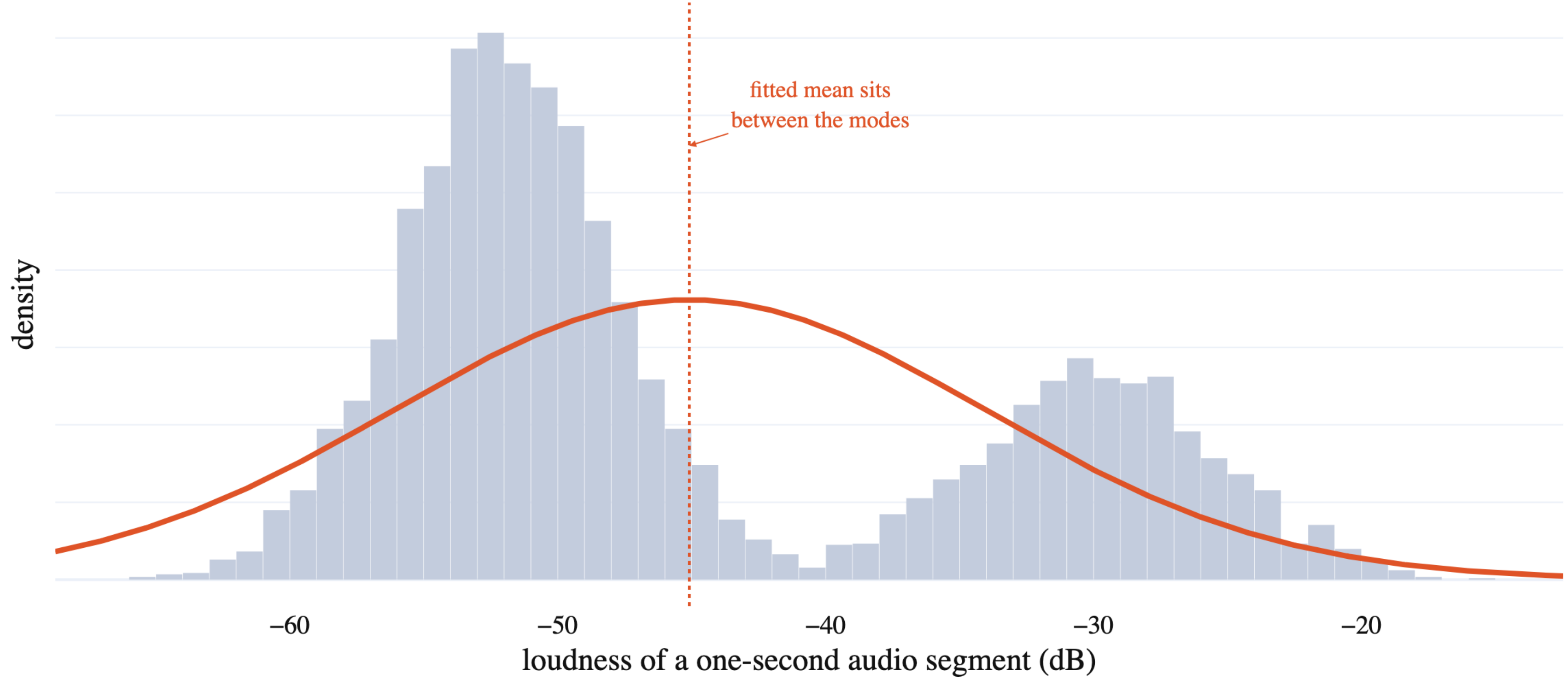
Often data has multiple modes, and a single Gaussian is not a great model for the distribution.



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Gaussian Mixture Models

- MAP Estimation
- Gaussians
  - Maximum Likelihood
  - Bias
- Gaussian Mixture Models

Questions

Gaussian Mixture Model (GMM)

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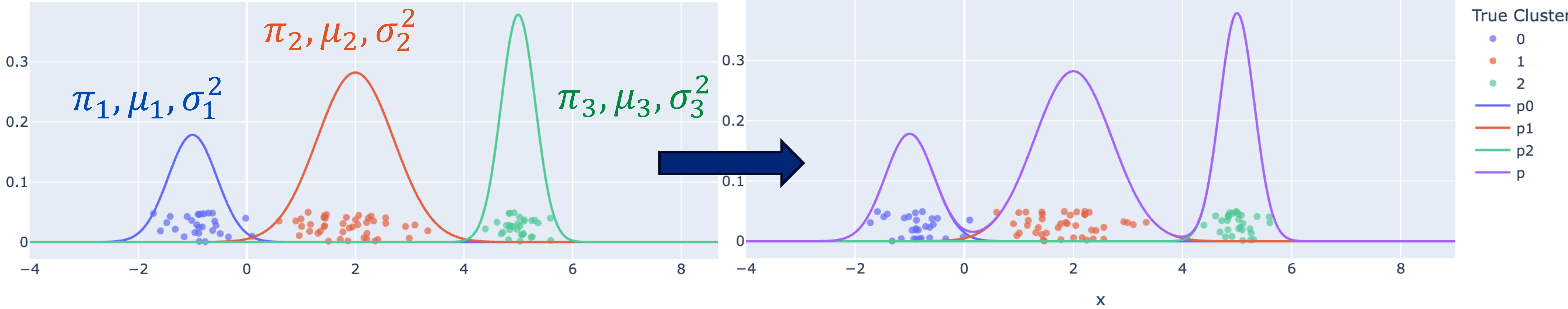
The Gaussian mixture model defines the probability of sampling data from a **weighted combination** of Gaussians

- **Example:** K=*3*



- Note the data doesn't have a color or a y-value: both were added for visualization purposes.

Gaussian Mixture Model (GMM)

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The Gaussian mixture model defines the probability of sampling data from a **weighted combination** of Gaussians:

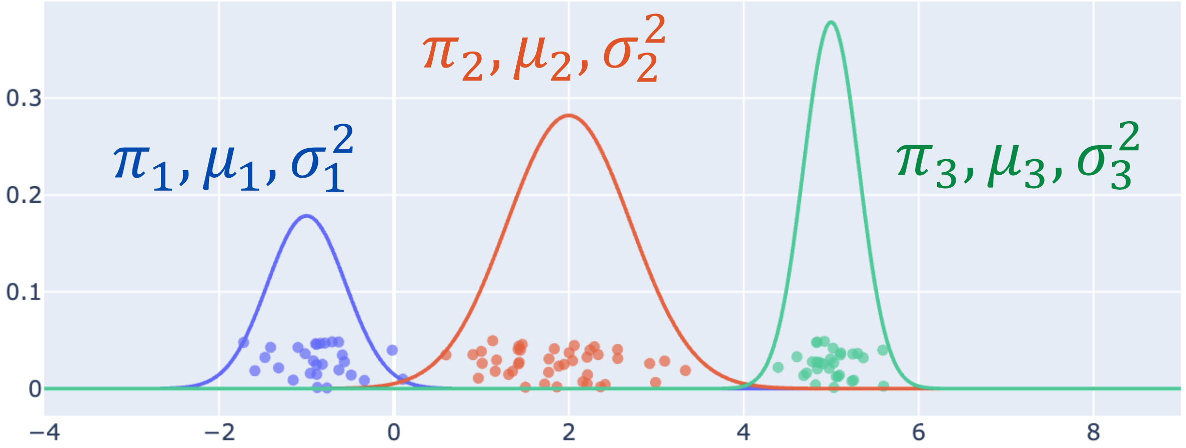
$$p(x \mid \pi, \mu_1, \sigma_{1,}^2 \dots, \mu_K, \sigma_K^2) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$$

where $\sum_{k=1}^{K} \pi_k = 1$ is a tabular probability distribution over clusters.

- **Example:** *K=3*

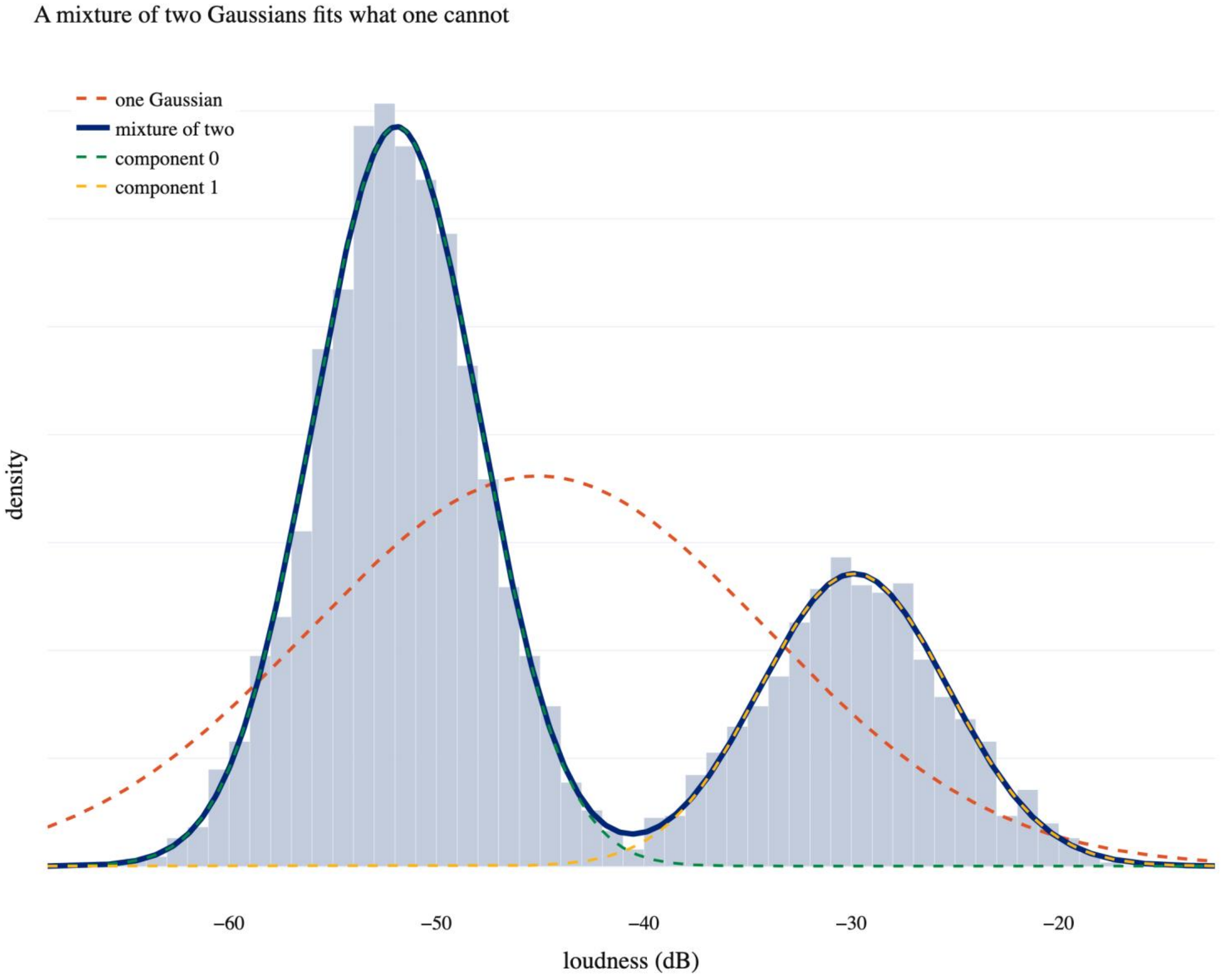
Parameters

```
pi = np.array([0.2, 0.5, 0.3])
mu = np.array([-1, 2, 5])
sigma = np.array([0.2, 0.5, .1])
```



Demo

Gaussian Mixture Model



The GMM is a **Latent Variable** Model

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We introduce a **latent** (unobserved) **assignment variable** z_n for each x_n.

$$p(z_n) = \pi_{z_n}$$

Think of this as the color in the earlier visualization

We then define the likelihood of x_n given z_n as:

Density for is based on it's Gaussian.

$$p(x_n \mid z_n) = \mathcal{N}\left(x_n \mid \mu_{z_n}, \sigma_{z_n}^2\right)$$

The joint probability of the x_n and z_n is then:

$$p(x_n, z_n) = p(x_n \mid z_n)\, p(z_n) = \mathcal{N}\left(x_n \mid \mu_{z_n}, \sigma_{z_n}^2\right)\pi_{z_n}$$

If we **marginalize** over the unknown z_n we obtain the **GMM**:

$$p(x_n) = \sum_{k=1}^{K} p(x_n, z_n = k) = \sum_{k=1}^{K} \mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\pi_k$$

x	z
1.3	1
9.6	3
2.4	1
5.7	2
…	…

The GMM is a **Generative Model**

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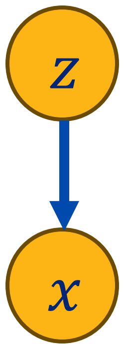
A **generative model** is a model that can **generate the data**.

$$p(x_n, z_n) = p(z_n)\, p(x_n \mid z_n)$$

We can construct **graph** where the parents of a variable are the variables in the **conditional distribution.**



- This is a **graphical model** and is used to represent the **conditional structure** of a probability model.

We can sample data from a graphical model using **ancestor sampling.**

- Sample all the variables that have no parents: $p(z_n)$
- Sample variables after all their parents have been sampled: $p(x_n \mid z_n)$

Example

Sample z_n from **Discrete**$(\pi = [.1, .3, .6])$
➔ $z_n = 2$

Sample x_n from $\mathcal{N}(x_n \mid \mu_2, \sigma_2^2)$
➔ $x_n = 4.8$

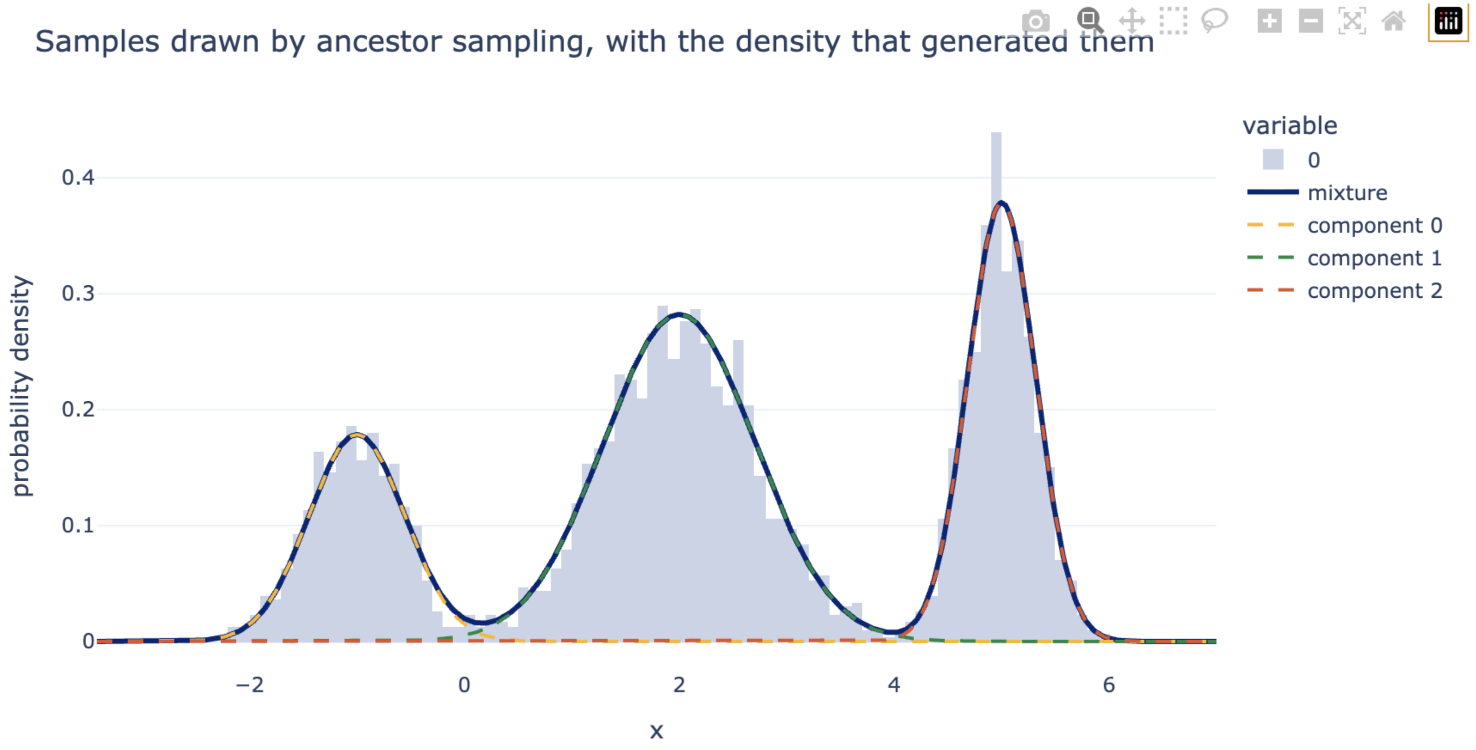
Demo

Sampling from a GMM

```
pi_true = np.array([0.2, 0.5, 0.3])
mu_true = np.array([-1.0, 2.0, 5.0])
var_true = np.array([0.2, 0.5, 0.1])

N = 3000
z = rng.choice(3, size=N, p=pi_true)                 # first the latent component
x = rng.normal(mu_true[z], np.sqrt(var_true[z]))     # then the observation

pd.Series(z).value_counts(normalize=True).sort_index().round(3)
```



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The Connection to K-Means

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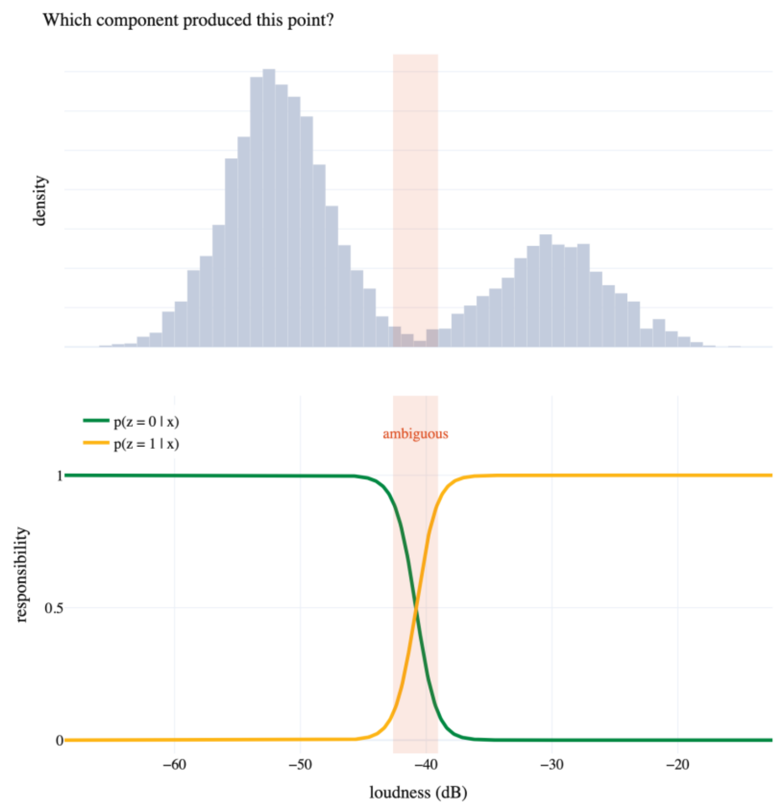
In Lecture 2, k-means assigned points to clusters using the same symbol $z_n \in \{1, \dots, K\}$ as today.

- k-means gives a **hard assignment**. Each point belongs to one cluster.

For a point midway between two clusters, that is a poor description.

A Gaussian mixture model keeps the same latent variable z_n but treats it as **random**.

- Instead of a single assignment, we obtain a distribution over the clusters.
- Instead of centers, each cluster has a mean, a variance, and a weight.



Latent Variable Posteriors

We are interested in modeling the **distribution** (*the uncertainty*) over the **cluster assignment** z.

- **Prior:** $p(z_n) = \pi_{z_n}$
- **Likelihood:** $p(x_n \mid z_n) = \mathcal{N}(x_n \mid \mu_z, \sigma_z^2)$

We can use Bayes rule to compute the posterior:

$$p\left(z_n = k \mid x_n, \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}\right) = \frac{p(x_n \mid z_n = k)p(z_n = k)}{\sum_{j=1}^{K} p(x_n \mid z_n = j)p(z_n = j)} = \frac{\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\pi_k}{\sum_{j=1}^{K} \mathcal{N}\left(x_n \mid \mu_j, \sigma_j^2\right)\pi_j}$$

If we **could estimate** $\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}$ **from the data** we could compute this distribution over cluster assignments for each data point.

We are interested in modeling the **distribution** (*the uncertainty*) over the **cluster assignment** z.

- **Prior:** $p(z_n) = \pi_{z_n}$
- **Likelihood:** $p(x_n \mid z_n) = \mathcal{N}(x_n \mid \mu_z, \sigma_z^2)$

We can use Bayes rule to compute the posterior:

$$p\left(z_n = k \mid x_n, \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}\right) = \frac{p(x_n \mid z_n = k)p(z_n = k)}{\sum_{j=1}^{K} p(x_n \mid z_n = j)p(z_n = j)} = \frac{\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\pi_k}{\sum_{k=1}^{K} \mathcal{N}\left(x_n \mid \mu_j, \sigma_j^2\right)\pi_j}$$

If we **could estimate** $\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}$ **from the data** we could compute this distribution over cluster assignments for each data point.

Trying the Stationary Point Method

Can we just apply the stationary point approach to maximize the likelihood?

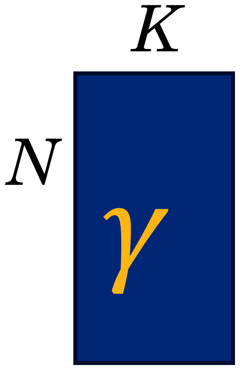
$$\arg\max_{\{\pi_k,\mu_k,\sigma_k^2\}_{k=1}^K} \sum_n \ln\left(\sum_k \pi_k \, \mathcal{N}(x_n \mid \mu_k, \sigma_k^2)\right)$$

Taking the derivative with respect to μ_k and applying the **calculus chain** rule:

$$\frac{\partial}{\partial \mu_k} \sum_n \ln\left(\sum_j \pi_j \, \mathcal{N}(x_n \mid \mu_j, \sigma_j^2)\right) = \sum_n \frac{\pi_k \, \mathcal{N}(x_n \mid \mu_k, \sigma_k^2)}{\sum_j \pi_j \, \mathcal{N}(x_n \mid \mu_j, \sigma_j^2)} \cdot \frac{x_n - \mu_k}{\sigma_k^2}$$

To make math cleaner lets define $\gamma_{nk} = \frac{\pi_k \mathcal{N}(x_n|\mu_k,\sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_n|\mu_j,\sigma_j^2)}$ and set the derivative equal to 0:

$$\sum_n \gamma_{nk} \frac{x_n - \mu_k}{\sigma_k^2} = 0 \;\Rightarrow\; \mu_k = \frac{\sum_n \gamma_{nk} \, x_n}{\sum_n \gamma_{nk}}$$



➢Unfortunately, γ_{nk} depends on μ_k and so **we can't solve for $\boldsymbol{\mu_k}$ analytically**.

Estimating the GMM Parameters

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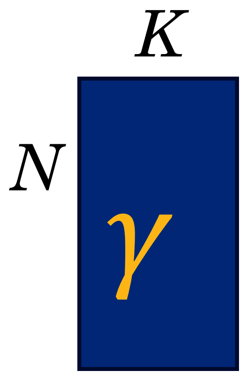
We could extend the model to include the **latent variable** z directly instead of marginalizing out their assignments:

$$\ln\left(p\left(\{x_n, z_n\}_{n=1}^N \mid \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^K\right)\right) = \sum_{n=1}^{N}\left(\ln(\pi_{z_n}) + \ln\left(\mathcal{N}\left(x_n \mid \mu_{z_n}, \sigma_{z_n}^2\right)\right)\right)$$

- We can now optimize the parameters directly (sum of log Gaussian terms).
- **But we don't know the values of $\{z_n\}_{n=1}^N$**

But, if we knew the parameters $\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^K$ than we could estimate $\{z_n\}_{n=1}^N$:

$$p\left(z_n = k \mid x_n, \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^K\right) = \frac{\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\pi_k}{\sum_{j=1}^K \mathcal{N}\left(x_n \mid \mu_j, \sigma_j^2\right)\pi_j} = \gamma_{nk}$$



Quick Recap

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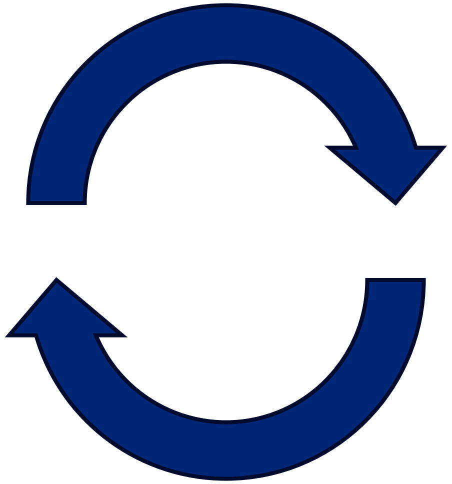
If we knew the model parameters, we could easily compute the cluster assignments.

Model Parameters

$$\left\{\pi_k, \mu_k, \sigma_k^2\right\}_{k=1}^{K}$$



Cluster Assignments

$$\{z_n\}_{n=1}^{N}$$

If we knew the cluster assignments, we could easily estimate the model parameters.

How can we solve this cyclic dependency?

Intuition Behind Expectation-maximization (EM)

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The **expectation-maximization algorithm** breaks the circle **by alternating, in the same way Lloyd's algorithm did for k-means** from Lecture 2.

Start with an initial guess for the parameters $\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}$:

Repeat until convergence:

- **E-step (Cluster Soft Assignment):** With the parameters fixed, compute the distribution over each point's **assignment using Bayes Theorem**.

$$p\left(z_n = k \mid x_n, \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}\right) = \frac{\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\pi_k}{\sum_{j=1}^{K} \mathcal{N}\left(x_n \mid \mu_j, \sigma_j^2\right)\pi_j} = \gamma_{nk}$$

- **M-step (Move Clusters).** With the assignment probabilities fixed, choose the parameters that **maximize the expected log likelihood**.

Each iteration **increases the log likelihood**, so the procedure **converges**.

The EM Algorithm: **E-step**

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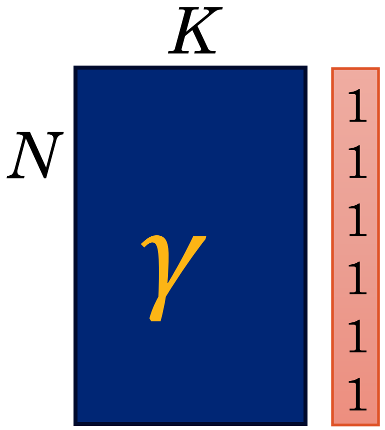
E-Step: We compute "soft" assignments for the unknown $\{z_n\}_{n=1}^N$

$$\gamma_{nk} = p\left(z_n = k \mid x_n, \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^K\right) = \frac{p(x_n \mid z_n = k)p(z_n = k)}{\sum_{j=1}^K p(x_n \mid z_n = j)p(z_n = j)}$$

$$= \frac{\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right) \pi_k}{\sum_{j=1}^K \mathcal{N}\left(x_n \mid \mu_j, \sigma_j^2\right) \pi_j}$$

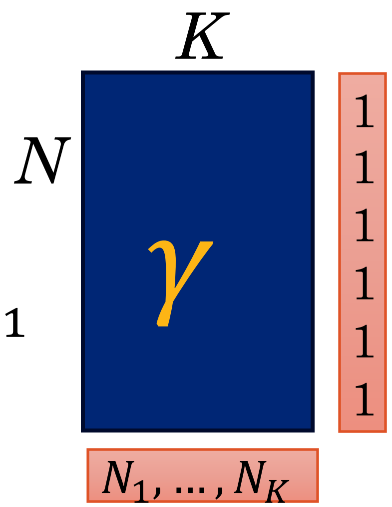
The soft assignments sum to $N = \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk}$ and we define the **cluster pseudo counts**:

$$N_k = \sum_{n=1}^N \gamma_{nk}$$



Cluster "pseudo counts" $N_1, \dots, N_K$

The EM Algorithm: **E-step**



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E-Step: We compute "soft" assignments for the unknown $\{z_n\}_{n=1}^{N}$

$$\gamma_{nk} = \frac{\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right) \pi_k}{\sum_{j=1}^{K} \mathcal{N}\left(x_n \mid \mu_j, \sigma_j^2\right) \pi_j}$$

Using the γ_{nk} we can compute the **expected log likelihood:**

$$Q\left(\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}\right) = \sum_{n=1}^{N} \sum_{k=1}^{K} \gamma_{nk} \ln\left(p\left(x_n, z_n = k \mid \pi, \{\mu_k, \sigma_k^2\}_{k=1}^{K}\right)\right)$$

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} \gamma_{nk} \left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\right)\right)$$

We can now **maximize** the expected log likelihood.

The EM Algorithm: M-Step (π_k)

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M-Step: Compute the parameters that **maximize** the expected log likelihood Q:

$$Q\left(\left\{\pi_k, \mu_k, \sigma_k^2\right\}_{k=1}^{K}\right) = \sum_{n=1}^{N} \sum_{k=1}^{K} \gamma_{nk} \left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\right)\right)$$

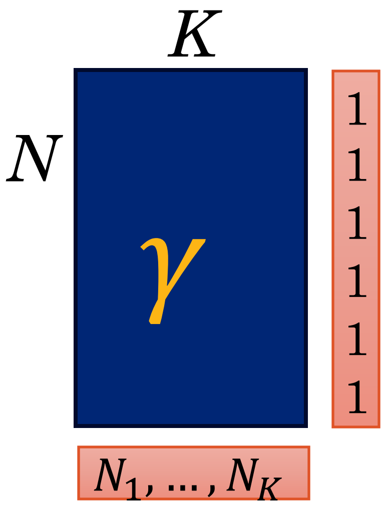
Method of Lagrange Multipliers (Lecture 3) for the **normalization constraint**

Optimizing with respect to π_k:

$$\frac{\partial}{\partial \pi_k}\left(\sum_{n=1}^{N} \sum_{k=1}^{K} \gamma_{nk} \left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\right)\right) - \lambda\left(\sum_{k=1}^{K} \pi_k - 1\right)\right)$$

$$= \sum_{n=1}^{N} \frac{\gamma_{nk}}{\pi_k} - \lambda = 0 \quad \Rightarrow \quad \pi_k = \frac{1}{\lambda} \sum_{n=1}^{N} \gamma_{nk} = \frac{N_k}{\lambda}$$

$$\boxed{N_k = \sum_{n=1}^{N} \gamma_{nk}}$$



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The EM Algorithm: M-Step (π_k)

M-Step: Compute the parameters that **maximize** the expected log likelihood Q:

$$Q\left(\left\{\pi_k, \mu_k, \sigma_k^2\right\}_{k=1}^{K}\right) = \sum_{n=1}^{N}\sum_{k=1}^{K} \gamma_{nk}\left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\right)\right)$$

Optimizing with respect to π_k:

$$\pi_k = \frac{1}{\lambda}\sum_{n=1}^{N}\gamma_{nk} = \frac{N_k}{\lambda}$$

Need figure out the **Lagrange Multiplier**

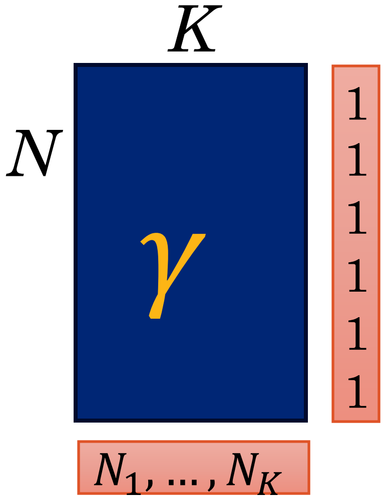
We know that $\sum_{k=1}^{K}\pi_k = 1$ so:

$$\sum_{k=1}^{K}\pi_k = \sum_{k=1}^{K}\frac{N_k}{\lambda} = \frac{N}{\lambda} = 1$$

therefore $\lambda = N$

$$\boxed{\pi_k = \frac{N_k}{N}}$$

Intuitively this is the proportion of points in cluster k



The EM Algorithm: **M-Step** (μ_k)

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M-Step: Compute the parameters that maximize Q:

$$Q\left(\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}\right) = \sum_{n=1}^{N}\sum_{k=1}^{K} \gamma_{nk}\left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\right)\right)$$

Optimizing with respect to μ_k:

$$\frac{\partial}{\partial \mu_k}\sum_{n=1}^{N}\sum_{k=1}^{K} \gamma_{nk}\left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \mid \mu_k, \sigma_k^2\right)\right)\right) = \sum_{n=1}^{N} \gamma_{nk}\, \sigma_k^{-2}(x_n - \mu_k) = 0$$

$$\sigma_k^{-2}\sum_{n=1}^{N} \gamma_{nk}\, x_n = \sigma_k^{-2}\mu_k \sum_{n=1}^{N} \gamma_{nk} = \sigma_k^{-2}\mu_k N_k$$

$$\mu_k = \frac{1}{N_k}\sum_{n=1}^{N} \gamma_{nk}\, x_n$$

The EM Algorithm: **M-Step** (σ_k^2)

9624861

M-Step: Compute the parameters that maximize Q:

$$Q\left(\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^K\right) = \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk} \left(\ln(\pi_k) + \ln\left(\mathcal{N}\left(x_n \,|\, \mu_k, \sigma_k^2\right)\right)\right)$$

Optimizing with respect to σ_k^2 follows the same stationary point derivation as the single Gaussian, now weighted by the responsibilities γ_{nk}:

$$\sigma_k^2 = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \left(x_n - \mu_k\right)^2$$

Summarizing the other parameters:

$$N_k = \sum_{n=1}^N \gamma_{nk}, \qquad \pi_k = \frac{N_k}{N}, \qquad \mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \, x_n$$

The EM Algorithm for GMMs

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1. **Initialize** the model parameters $\{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}$) often using k-means)
2. **Iterate** until convergence
   - **E-Step:** compute expected log-likelihood

$$\gamma_{nk} = \frac{\mathcal{N}(x_n \mid \mu_k, \sigma_k^2)\pi_k}{\sum_{j=1}^{K} \mathcal{N}(x_n \mid \mu_j, \sigma_j^2)\pi_j}$$

   - **M-Step:** maximize the expected log-likelihood

$$N_k = \sum_{n=1}^{N} \gamma_{nk}, \qquad \pi_k = \frac{N_k}{N}, \qquad \mu_k = \frac{1}{N_k}\sum_{n=1}^{N} \gamma_{nk}\, x_n$$

$$\sigma_k^2 = \frac{1}{N_k}\sum_{n=1}^{N} \gamma_{nk}\,(x_n - \mu_k)^2$$

The EM Algorithm

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The **Expectation-Maximization (EM)** algorithm is a standard technique for fitting **latent variable** models like the Gaussian Mixture Model.

1. **Initialize** the model parameters θ.
2. **Iterate** until convergence.
   - **E-Step:** Compute expected log-likelihood.

Easy to optimize **joint probability**

Current distribution over the latent Z

$$Q(\theta') = \sum_{Z} \ln(p(X, Z \mid \theta'))\, p(Z \mid X, \theta)$$

   - **M-Step:** Maximize the expected log-likelihood.

$$\theta = \arg\max_{\theta'} Q(\theta')$$

Updates distribution over Z.

Convergence and Local Optima

9624861

Except for the **variance collapse failure modes**, the likelihood is bounded from above and the EM Algorithm **only increases** the likelihood with each change in parameters:

- typically to a **local** maximum, not necessarily the global one.

Initialization matters.

- Different starting points can land on different local optima
- Run k-means first: use its centers for the initial μ_k, the whole-dataset variance for σ_k^2, and $\frac{1}{K}$ for π_k (default for sklearn's GaussianMixture model).

A failure mode specific to mixtures: if a component's mean lands on a single data point and $\sigma_k^2 \to 0$, that term's likelihood diverges.

- Often set a variance floor (or a MAP/Bayesian prior penalizing tiny variances) and sensible initialization; (usually enough)

Convergence, Initialization, and Local Optima

9624861

Each EM iteration increases the log likelihood, and the likelihood is bounded above, so the algorithm converges.

It converges to a **local** maximum. Different starting points give different answers, exactly as Lloyd's algorithm did in Lecture 2.

Initialization matters.

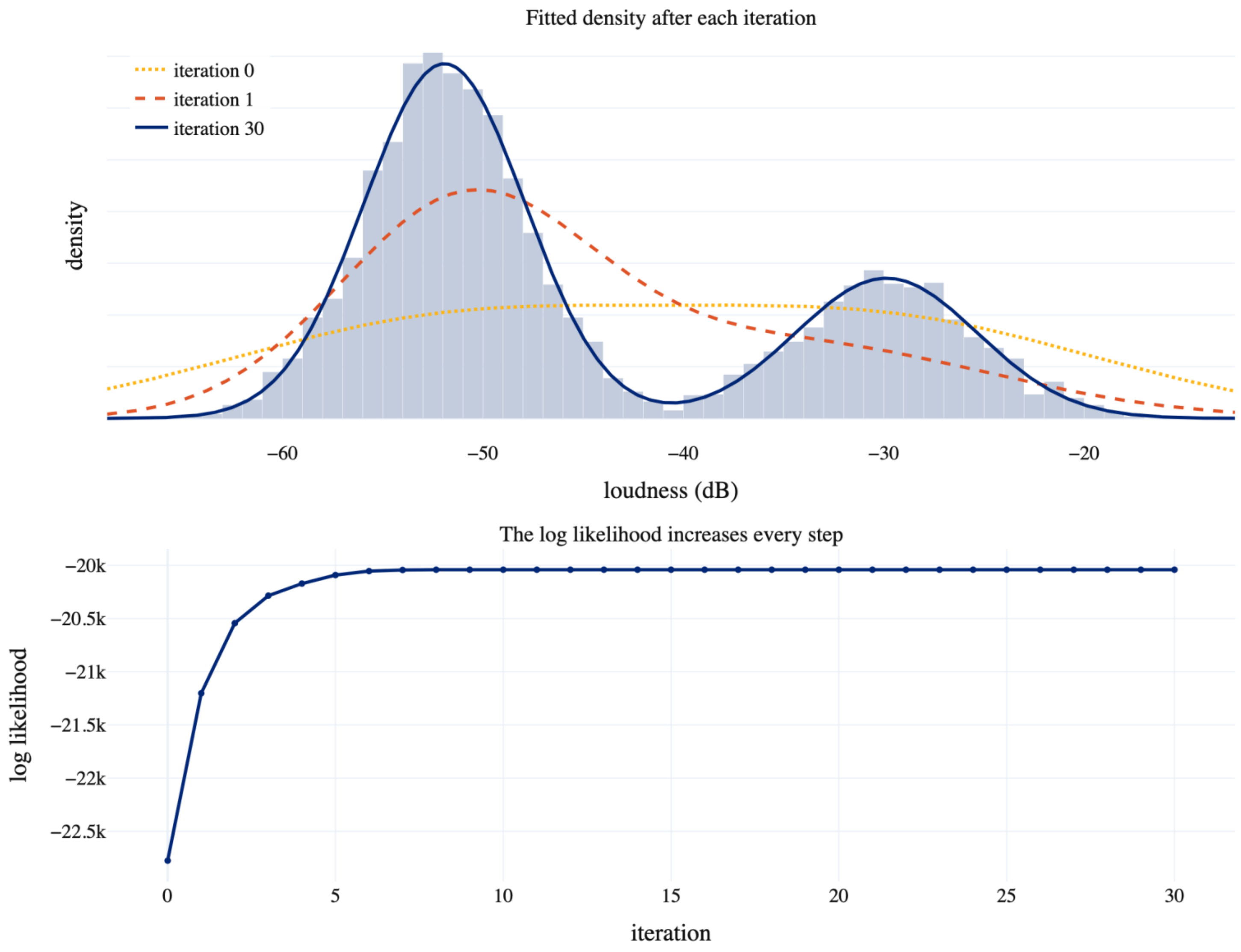
- Running k-means first and using its centers as the initial μ_k is standard practice.
- The initial σ_k^2 are often set to the variance of the whole dataset, and the π_k to $1/K$.
- sklearn.mixture.GaussianMixture defaults to k-means initialization and n_init=1.

A failure mode specific to mixtures. If one component collapses onto a single point, its variance σ_k^2 shrinks toward zero and the likelihood diverges.

- The likelihood is unbounded, so a global maximum does not exist in any useful sense.
- Implementations add a small constant to σ_k^2 to prevent it.

Demo

Implementing EM for GMMs



What We Did Today

9624861

Fitting a Gaussian. Maximum likelihood returns the sample mean and the sample variance, which depend on the data only through $\sum_n x_n$ and $\sum_n x_n^2$.

Bias. The maximum likelihood variance is low by a factor of $\frac{N-1}{N}$, because deviations are measured from a mean estimated on the same data.

Mixture models. A weighted sum of Gaussians describes data that a single Gaussian cannot. The component that generated each point is a latent variable z_n.

Why EM is necessary. The log likelihood of a mixture puts a sum inside the logarithm, so the stationary point method yields no closed-form solution.

EM. Alternate between computing the responsibilities γ_{nk} by Bayes' theorem and maximizing the expected complete-data log likelihood. Each step is closed form; together they converge to a local maximum.

Lecture 5

Density Estimation and Gaussian Mixture Models

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters:

- **Probability:** Chapter 2.[1-2], 2.3.[1-3] (we will return to 2.4 onward later)
- **Distributions:** Chapter 3.1, 3.2.1 (the rest of 3 is more advanced Gaussians)
- **Clustering:** Chapter 15.1 (k-means), 15.2 (Gaussian Mixture Models)